The difference in the phases between the retarded (deviated) and un-retarded (non-deviated) light rays is about ¼ of original wave length (i.e., λ/4). Human eyes cannot detect these minute changes in the phase of light. The phase contrast microscope has special devices such as annular diaphragm and phase plate, which convert these minute phase changes into brightness (amplitude) changes, so that a contrast difference can be created in the final image. This contrast difference can be easily detected by human eyes.

In phase contrast microscope, to get contrast, the diffracted waves have to be separated from the direct waves. This separation is achieved by the sub-stage annular diaphragm.

The annular diaphragm illuminates the specimen with a hollow cone of light. Some rays (direct rays) pass through the thinner region of the specimen and do not undergo any deviation and they directly enter into the objective lens. The light rays passing through the denser region of the specimen get regarded and they run with a delayed phase than the non-deviated rays. Both the deviated and non deviated light has to pass through the phase plate kept on the back focal plane of the objective to form the final image. The difference in phase (Wavelength) gives the contrast for clear visibility of the object. Figure 2.4 Microscopic image comparing phase and bright field microscopy.

Infobits

Whenever light (or any wave in general) goes from one medium to another, some of the energy of the wave is "reflected" back through the first medium cut the same angle as the incident wave and some of the energy is refracted (bent). Through the second medium when light goes from a low refractive index medium to a high refractive index medium such as air to water the reflection undergoes a 180° phase change. Light waves that are in phase (that is, their peaks and valleys exactly coincide) reinforce one another and their total intensity increases.

Light waves that are out of phase by exactly one-half wavelength cancel each other and result in no intensity. That is darkness wavelengths that are out of phase by any amount will produce some degree of cancellation and result in brightness less than maximum, but more than darkness. Thus, contrast is provided by differences is light intensity that result from differences in refractive indices in parts of the specimen that put light waves indices in parts of the specimen that put light waves more or less out of phase. As a result, the specimen appears as various levels of darks against a bright background.

2.1.4 Applications

- Phase contrast microscope enables the visualization of unstained living cells.
- It makes highly transparent objects more visible.

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